DL RAINFALL PIPELINE — SETUP GUIDE

Data Setup Guide

Step-by-step instructions for downloading the three external datasets required to run the full pipeline — ERA5, NASA IMERG, and climate indices. Raw station rainfall data and pre-processed CSVs are handled separately.

ERA5 REANALYSIS
~2–5 GB, 11 files
▲ Download required

NASA IMERG
~500 MB, 129 files
▲ Download required

CLIMATE INDICES (ONI / DMI)
Already in repo
✓ No action needed

PROCESSED CSVS
All 96 stations
✓ Already in repo

1 ERA5 Weather Reanalysis Data

ERA5 is ECMWF's global atmospheric reanalysis — it provides daily temperature, humidity, wind, pressure, and rainfall estimates at ~28 km resolution. It is the primary atmospheric feature source for the peninsular rivers (Johor, Kedah, Klang, Kuantan).

The bounding box below (**[8, 99, 0, 120]**) covers all of Malaysia including Sabah and Sarawak, so this single download works for all 96 stations.

1 Register for a free CDS account

Go to **https://cds.climate.copernicus.eu** → click **Register** → create a free account. After logging in, go to your profile and copy your **API Key**.

2 Install the CDS API client

```
pip install cdsapi
```

3 Create the credentials file

Create a file named `.cdsapirc` in your home directory:

- **Windows:** `C:\Users\<your-name>\.cdsapirc`
- **Mac / Linux:** `~/.cdsapirc`

File contents:

```
url: https://cds.climate.copernicus.eu/api
key: <paste-your-API-key-here>
```

4

Run the download script

Create folder `data/era5/` in the project root, then run:

```
import cdsapi
from pathlib import Path

client = cdsapi.Client()
output_dir = Path("data/era5")
output_dir.mkdir(exist_ok=True)

VARIABLES = [
    "2m_temperature",
    "2m_dewpoint_temperature",
    "surface_pressure",
    "10m_u_component_of_wind",
    "10m_v_component_of_wind",
    "total_column_water_vapour",
    "total_precipitation",
]

for year in range(2015, 2026):
    out_file = output_dir / f"era5_malaysia_{year}.nc"
    if out_file.exists():
        print(f"Already exists: {out_file.name}, skipping.")
        continue
    print(f"Downloading {year}...")
    client.retrieve(
        "reanalysis-era5-single-levels",
        {
            "product_type": "reanalysis",
            "variable": VARIABLES,
            "year": str(year),
            "month": [f"{m:02d}" for m in range(1, 13)],
            "day":   [f"{d:02d}" for d in range(1, 32)],
            "time":  "12:00",
            "area":  [8, 99, 0, 120],  # [N, W, S, E] — all of Malaysia
            "format": "netcdf",
        },
        str(out_file),
    )
    print(f"Saved: {out_file.name}")
```

Expected output: 11 files — `era5_malaysia_2015.nc` through `era5_malaysia_2025.nc` in `data/era5/` . Each request is queued — allow 10–30 minutes per year. The script skips files that already exist so it is safe to re-run after interruption.

> **ERA5 for East Malaysia (Padas / Sarawak):** The bounding box above covers Sabah (~5°N, 116°E) and Sarawak (~1.5°N, 110°E). The Padas and Sarawak models in this project used IMERG + climate indices instead (because the ERA5 files on hand only covered Pahang), but a full Malaysia ERA5 download gives you the option to use ERA5 for those rivers in future work.

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2 NASA IMERG Monthly Satellite Rainfall

IMERG (Integrated Multi-satellitE Retrievals for GPM) is NASA's global monthly rainfall product at 0.1° resolution (~11 km). It provides finer-scale precipitation estimates than ERA5 and has global coverage — making it the primary precipitation feature for Padas and Sarawak (where ERA5 was not available during this project).

1 Register for NASA Earthdata

Go to **https://urs.earthdata.nasa.gov** → click **Register** → create a free account.

After registering, go to **Applications** → **Authorized Apps** and add **"NASA GESDISC DATA ARCHIVE"** — this authorisation is required for the download to succeed.

2 Create the credentials file

Create a file named `.netrc` in your home directory:

- **Mac / Linux:** `~/.netrc`
- **Windows:** `C:\Users\<your-name>\.netrc` *and* `C:\Users\<your-name>\_netrc`

```
machine urs.earthdata.nasa.gov
    login <your-earthdata-username>
    password <your-earthdata-password>
```

3 Run the download script

Create folder `data/imerg_monthly/` , then run:

```
import requests, calendar
from pathlib import Path

output_dir = Path("data/imerg_monthly")
output_dir.mkdir(exist_ok=True)

BASE_URL = "https://gpm1.gesdisc.eosdis.nasa.gov/data/GPM_L3/GPM_3IMERGM.07"

for year in range(2015, 2026):
    for month in range(1, 13):
        if year == 2025 and month > 9:  # V07B available to ~Sep 2025
```

```
            break
        filename = (
            f"3B-MO.MS.MRG.3IMERG.{year}{month:02d}01"
            f"-S000000-E235959.{year}{month:02d}.V07B.HDF5"
        )
        url = f"{BASE_URL}/{year}/{filename}"
        out = output_dir / filename
        if out.exists():
            print(f"Already exists: {filename}, skipping.")
            continue
        print(f"Downloading {filename}...")
        r = requests.get(url, stream=True)
        if r.status_code == 200:
            with open(out, "wb") as f:
                for chunk in r.iter_content(chunk_size=8192):
                    f.write(chunk)
            print(f"  Saved ({out.stat().st_size // 1024} KB)")
        else:
            print(f"  ERROR {r.status_code}: {url}")
```

> **Expected output:** 129 HDF5 files (Jan 2015 – Sep 2025) in `data/imerg_monthly/`, ~500 MB total. The script skips existing files — safe to re-run.

> **If you get a 401 error:** Your Earthdata account has not authorised GES DISC yet. Go back to Step 1 → Applications → Authorized Apps and make sure "NASA GESDISC DATA ARCHIVE" is in the list. Also check that your `.netrc` / `_netrc` file is saved correctly with no extra spaces.

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3 Climate Indices — ONI & DMI

> ✓ **Already included in the repo.** The files `data/processed/climate_indices_daily.csv` and `data/processed/climate_indices_monthly.csv` are committed and ready to use. You do not need to download anything unless you want to extend coverage beyond 2026.

The two climate indices used in this project:

INDEX	FULL NAME	WHAT IT MEASURES	EFFECT ON MALAYSIA
ONI	Oceanic Niño Index	Sea surface temperature anomaly in the central Pacific (Niño 3.4 region)	Positive (El Niño) → less rain; Negative (La Niña) → more rain, stronger northeast monsoon
DMI	Dipole Mode Index (Indian Ocean Dipole)	Temperature difference between western and eastern Indian Ocean	Positive IOD → reduced moisture advection from Indian Ocean → less rain; Negative IOD → enhanced rain

TO UPDATE IN THE FUTURE (EXTENDING PAST 2026)

1 **Update ONI — from NOAA**

Source: **https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt**

```
import pandas as pd

oni = pd.read_csv(
    "https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt",
    sep=r"\s+", skiprows=1,
    names=["year","jan","feb","mar","apr","may","jun",
           "jul","aug","sep","oct","nov","dec"]
)
# Reshape to monthly rows, then merge into climate_indices_daily.csv
```

2 **Update DMI — from JAMSTEC**

Source: **https://www.jamstec.go.jp/aplinfo/sintexf/iod/Data/dmi.monthly.txt**

```
dmi = pd.read_csv(
    "https://www.jamstec.go.jp/aplinfo/sintexf/iod/Data/dmi.monthly.txt",
    sep=r"\s+", comment="#"
)
```

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4 Raw Station Rainfall Data

> ✓ **Processed CSVs already in repo.** All six rivers are pre-processed and committed. You only need the raw Excel files if you want to re-run data cleaning from scratch.

The raw data is provided by YBTM/JPS (Department of Irrigation and Drainage Malaysia) and is not publicly downloadable. Place the files at:

`data/raw/Permohonan Data YBTM - KE/Permohonan Data YBTM - KE/`

FILE	RIVER	STATE	STATIONS	FORMAT
`RF Sg. Johor 1.5.2015 - 1.6.2026.xlsx`	Sg. Johor	Johor	18	Incremental
`RF Sg. Kedah 1.5.2015 - 1.6.2026.xlsx`	Sg. Kedah	Kedah	11	Incremental
`RF Sg. Klang 1.5.2015 - 1.6.2026.xlsx`	Sg. Klang	Selangor / KL	19	Incremental
`RF Sg. Kuantan 1.5.2015 - 1.6.2026.xlsx`	Sg. Kuantan	Pahang	5	Incremental
`RF Sg. Padas 1.5.2015 - 1.6.2026.xlsx`	Sg. Padas	Sabah	3	**Cumulative**

				counter
`RF Sg. Sarawak 1.5.2015 - 1.6.2026.xlsx`	Sg. Sarawak	Sarawak	40	**Cumulative counter**

Padas and Sarawak use a different data format. These two files record a cumulative running total, not per-interval rainfall. Applying a direct sum to them produces thousands of mm per day. The notebook `01b_data_engineering_multiriver.ipynb` handles this automatically using `diff()` -based aggregation with a 50 mm/15-min spike cap. Do not process these files the same way as the peninsular rivers.

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5 Verify Your Setup

After completing the downloads above, your `data/` folder should look like this:

```
data/era5/                          ← download required (Section 1)
│   era5_malaysia_2015.nc
│   era5_malaysia_2016.nc  ... era5_malaysia_2025.nc
data/imerg_monthly/                  ← download required (Section 2)
│   3B-MO.MS.MRG.3IMERG.20150101-...201501.V07B.HDF5
│   3B-MO.MS.MRG.3IMERG.20150201-...201502.V07B.HDF5
│   ...  (129 files total, up to 202509)
data/processed/                      ← already in repo ✓
│   johor_daily_raw.csv          (18 stations)
│   kedah_daily_raw.csv          (11 stations)
│   klang_daily_raw.csv          (19 stations)
│   kuantan_daily_raw.csv        (5 stations)
│   padas_daily_raw.csv          (3 stations — East Malaysia)
│   sarawak_daily_raw.csv        (40 stations — East Malaysia)
│   imerg_stations_monthly.csv   (96 stations × 129 months)
│   climate_indices_daily.csv
│   climate_indices_monthly.csv
│   all_rivers_station_meta.json (lat/lon for all 96 stations)
data/raw/                             ← from YBTM/JPS
    Permohonan Data YBTM - KE/
        RF Sg. Johor 1.5.2015 - 1.6.2026.xlsx
        RF Sg. Kedah 1.5.2015 - 1.6.2026.xlsx
        RF Sg. Klang 1.5.2015 - 1.6.2026.xlsx
        RF Sg. Kuantan 1.5.2015 - 1.6.2026.xlsx
        RF Sg. Padas 1.5.2015 - 1.6.2026.xlsx
        RF Sg. Sarawak 1.5.2015 - 1.6.2026.xlsx
```

Once `data/era5/` and `data/imerg_monthly/` are populated, all scripts in `scripts/` and all notebooks will run without modification — paths are already configured.

6 Python Dependencies

A Using the project virtual environment (recommended)

```
# Windows
.venv_tf\Scripts\activate
pip install -r requirements.txt
```

B Or install individually

```
pip install tensorflow scikit-learn xgboost pandas numpy \
            h5py cdsapi requests netCDF4 openpyxl
```

PACKAGE	USED FOR
`tensorflow`	TCN model training (Phases 5–6)
`scikit-learn`	RidgeCV, StandardScaler (Phases 6–7)
`xgboost`	XGBoost Hurdle model (Phase 4)
`h5py`	Reading NASA IMERG HDF5 files
`netCDF4`	Reading ERA5 .nc files
`cdsapi`	Downloading ERA5 from Copernicus CDS
`requests`	Downloading IMERG from NASA GES DISC